

Food Component Detection for Dietary Recommendation ——Children's Fun Nutrition Guide

COMP4913 Capstone Project Presentation (Final)

kids-nutrition-fun.com

Student Name: LIU Yibo

Student ID No.: 22098545D

Supervisor: Dr. PANG Wai Man

Co-Examiner: Dr. ZHANG Mingjin

2nd Assessor: Prof. CHENG Ran

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1.Problem & Motivation

Why do children need a dedicated nutrition platform?

25%

of children aged 7–9 in Europe are overweight or obese (WHO COSI 2022–2024)



MyFitnessPal: Calorie Counter

MyFitnessPal, Inc.
包含广告·应用内购买商品

Existing apps are adult-only

MyFitnessPal, Cronometer rely on calorie dashboards & text-heavy interfaces — not suitable for children aged 3–12

No child-centred solution

No major platform combines AI personalisation, gamification, and evidence-based recipes in one bilingual tool

Critical development window

Ages 3–12 are formative for lifelong dietary habits — early, engaging intervention matters most

2.Project Objectives & Key Features

kids-nutrition-fun.com

Nutri AI Chatbot

Bilingual EN/ZH
Powered by GPT-4o Mini

Recipe System

50+ recipes
7 meal categories

Mini-Games

Nutrition Runner
Plate Builder

User System

Registration
& authentication

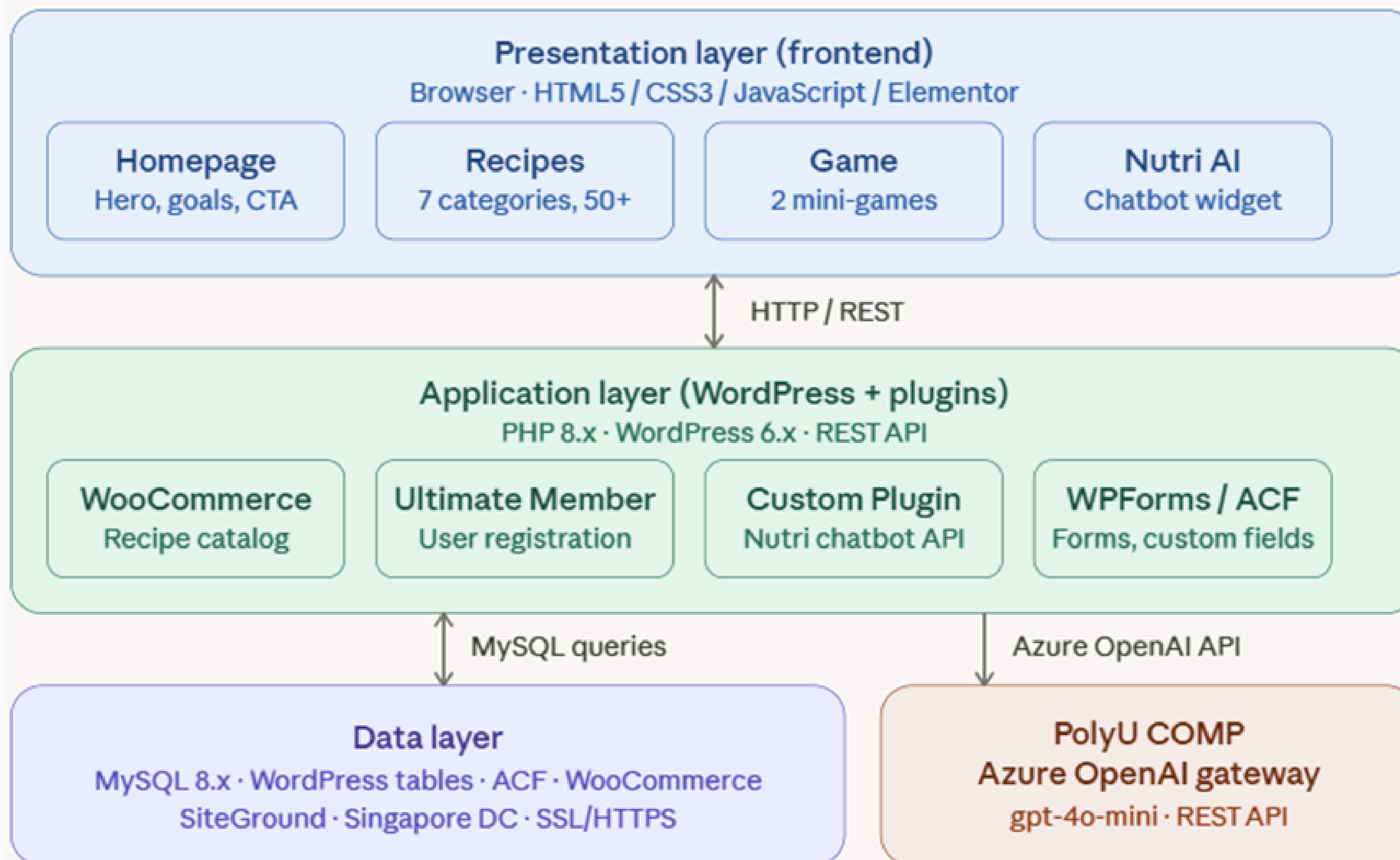
Contact System

WPForms inquiry
24–48h response

All running on WordPress 6.x + PHP 8.x · SiteGround Singapore · MySQL 8.x · Let's Encrypt SSL

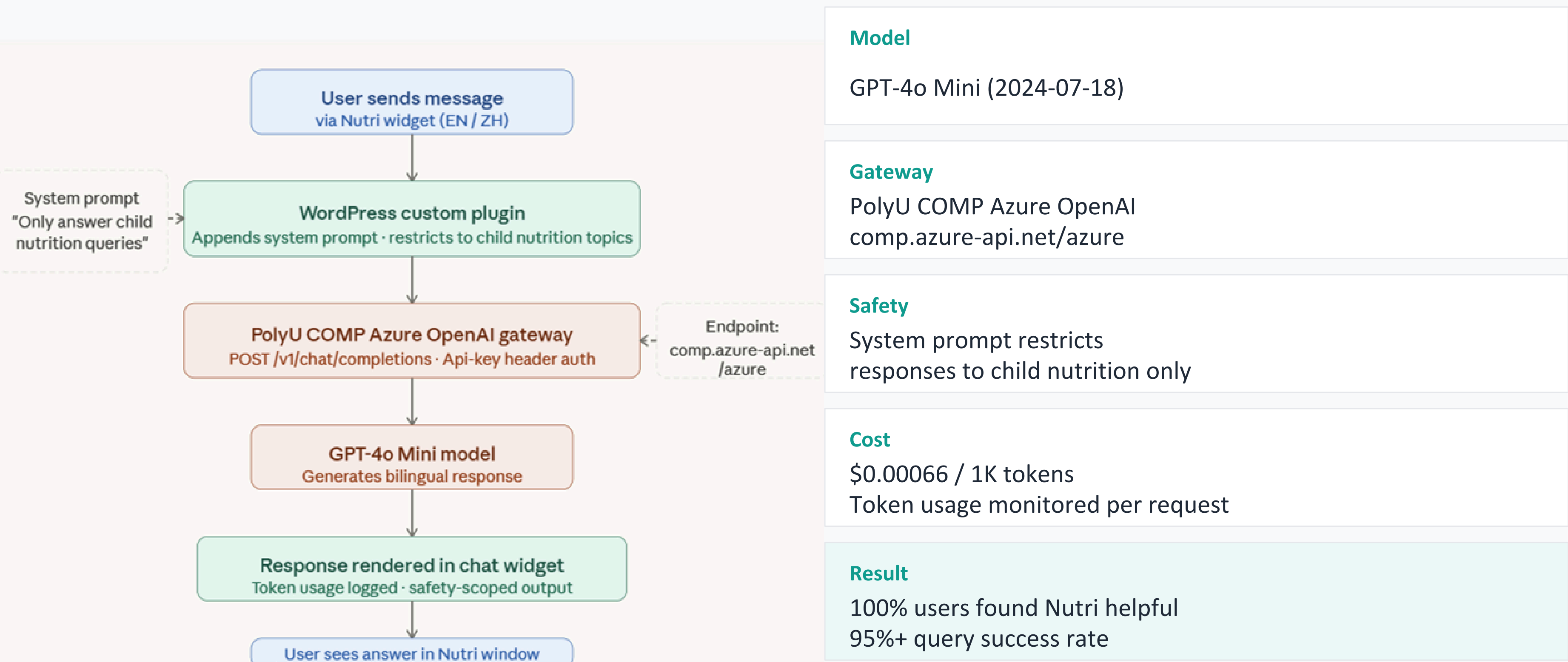
3.System Architecture

Three-tier web architecture



4.Key Feature 1 — Nutri AI Chatbot

Bilingual AI nutritional assistant powered by PolyU COMP Azure OpenAI



4. Key Feature 1 — Nutri AI Chatbot

Bilingual AI nutritional assistant powered by PolyU COMP Azure OpenAI

```
$system_prompt = "You are Nutri, a friendly bilingual (English & Chinese) nutrition assistant for Kids Nutrition Fun (https://kids-nutrition-fun.com/),  
// helping parents raise healthy children aged 3-12.  
Answer questions about children's nutrition, recipes, food allergies, healthy eating habits  
Always reply in the SAME language the user writes in (Chinese or English)  
Follow WHO & CDC guidelines  
Keep answers practical and under 150 words  
Recommend website features: personalized recipes, fun games, progress reports, free signup at https://kids-nutrition-fun.com/login";
```

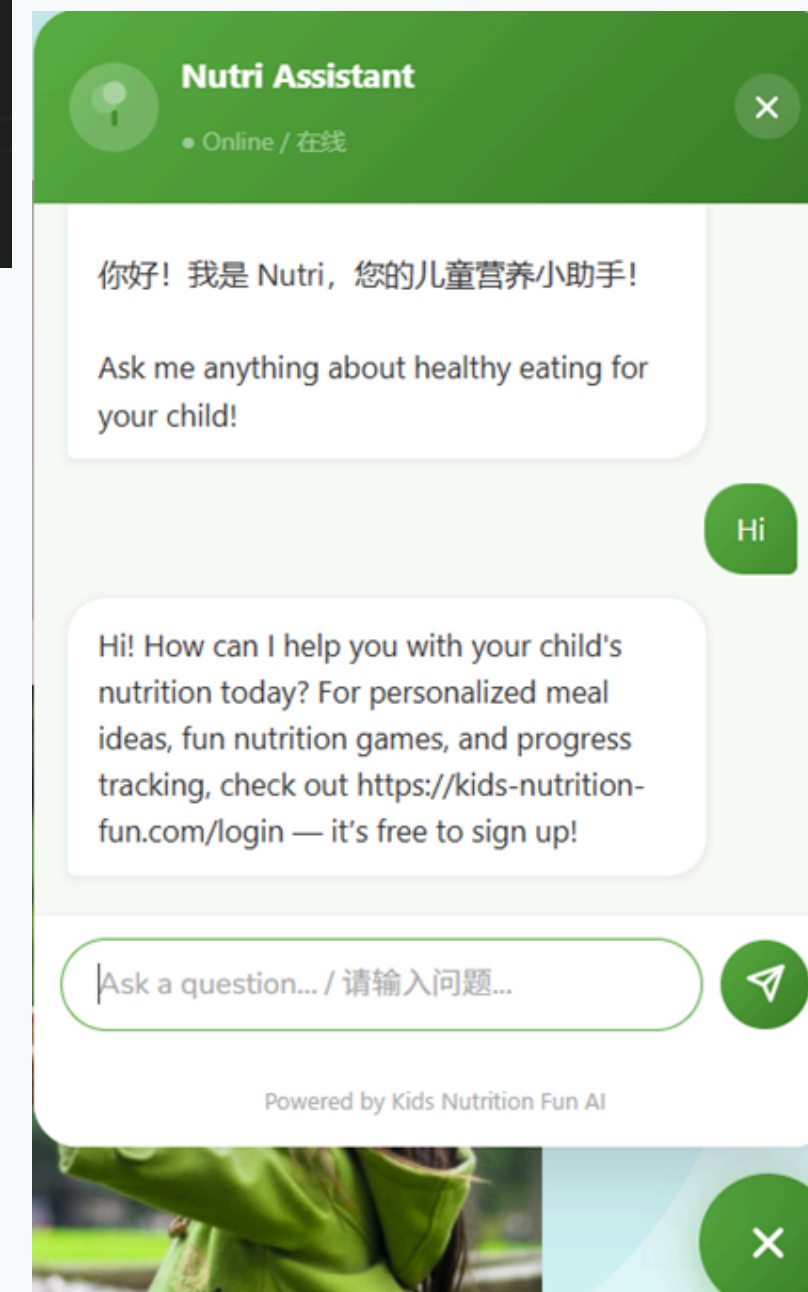
```
$messages = [['role' => 'system', 'content' => $system_prompt]];  
if (is_array($history)) {  
    foreach (array_slice($history, -6) as $h) {  
        if (isset($h['role'], $h['content'])) {  
            $messages[] = ['role' => sanitize_text_field($h['role']), 'content' => sanitize_text_field($h['content'])];  
        }  
    }  
}
```

System_prompt

- You are Nutri, a friendly bilingual (English & Chinese) nutrition assistant for Kids Nutrition Fun (<https://kids-nutrition-fun.com/>), helping parents raise healthy children aged 3-12.
- Answer questions about children's nutrition, recipes, food allergies, and healthy eating habits
- Always reply in the SAME language the user writes in (Chinese or English)
- Follow WHO & CDC guidelines
- Keep answers practical and under 150 words

100% users found Nutri helpful• 95%+ query success rate (n=20)

English Query



Chinese Query



5.Key Feature 2 — Recipe System

WooCommerce repurposed as a child-friendly recipe catalog

Recipes

Breakfast (8)

Dinner (9)

Lunch (8)

Salads & Bowls (9)

Snacks & Treats (8)

Soups & Stews (8)

Veggie Mains (8)



Dinner

Asian Vegetable Rice Noodle Bowl



Lunch

Awesom

7 Meal Categories

Breakfast

Dinner

Salads & Bowls

Veggie Mains

Lunch

Snacks & Treats

Soups & Stews



Home / Dinner / Asian Vegetable
Rice Noodle Bowl

Dinner, Veggie Mains

Asian Vegetable Rice Noodle Bowl

Categories: Dinner, Veggie Mains

- 50+ child-friendly recipes total
- Each recipe: photo · ingredients · steps · nutrition panel
- Filter by category · sort by name or latest
- Allergen warnings on every recipe

8/ 14

Key Feature 2 — Recipe System

Database design behind the WooCommerce-powered recipe catalog

Database ERD

USER		
int	user_id	PK
string	username	
string	email	
string	password_hash	
datetime	registered_at	

RECIPE		
int	recipe_id	PK
string	title	
string	category	
int	calories	
int	protein_g	
int	carbs_g	
int	fat_g	

RECIPE_LOG		
int	log_id	PK
int	user_id	FK
int	recipe_id	FK
date	logged_date	

RECIPE_CATEGORY		
int	category_id	PK
int	recipe_id	FK
string	category_name	

RECIPE_ALLERGEN		
int	allergen_id	PK
int	recipe_id	FK
string	allergen_name	

Entity Logic

USER

Single unified account model
(parent/guardian registers; no child sub-account hierarchy)

RECIPE

Core content entity — stores title, category, calories, macros
(protein / carbs / fat)

RECIPE_LOG

Bridge table — records which user viewed/saved which recipe
(One-to-Many)

RECIPE_CATEGORY

Maps each recipe to one of 7 meal types:
Breakfast · Lunch · Dinner · Snacks · Salads · Soups · Veggie

RECIPE_ALLERGEN

Stores per-recipe allergen tags to power dietary screening and
safe filtering

6.Key Feature 3 — Mini-Games

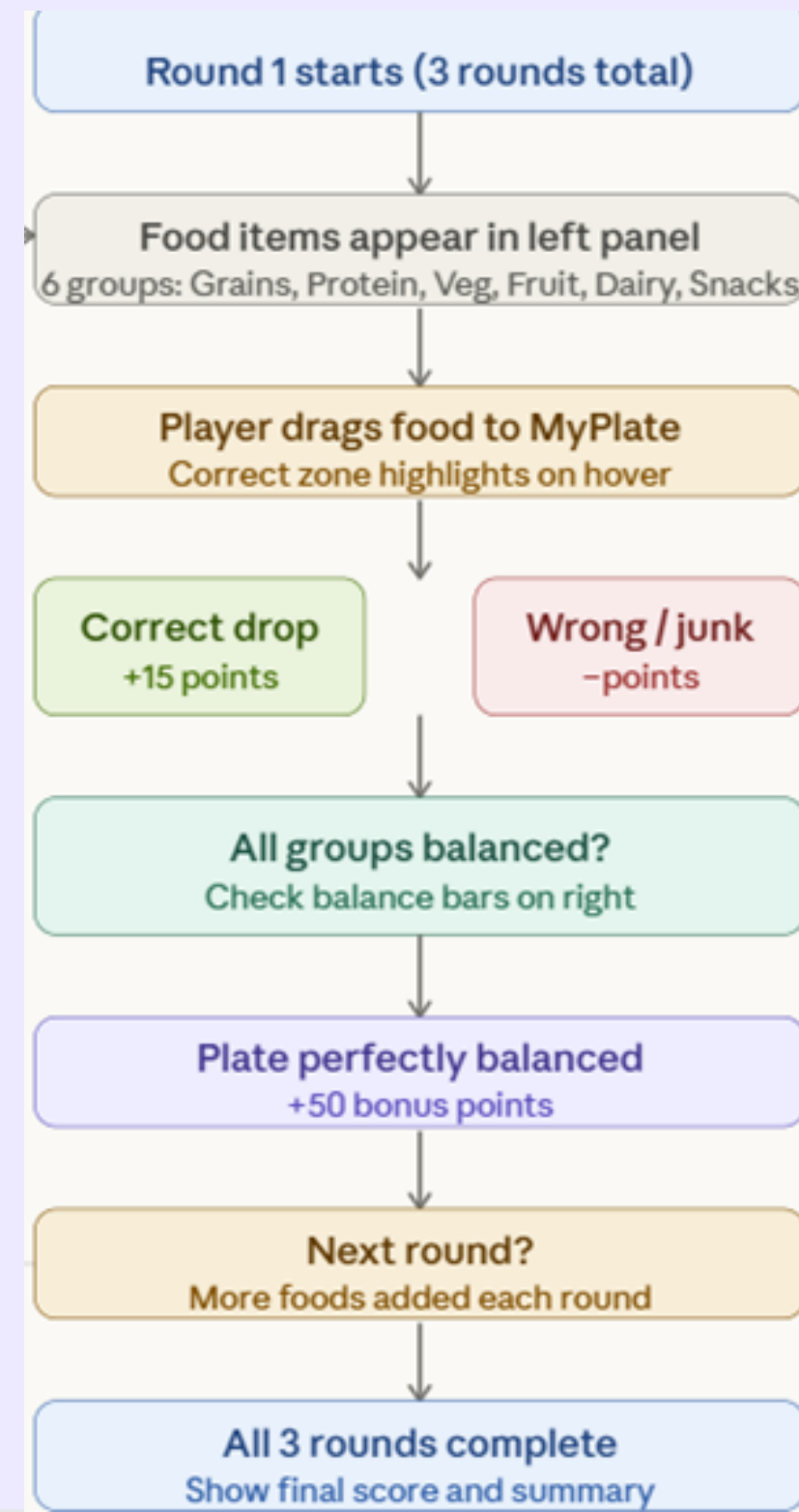
HTML files uploaded to SiteGround · embedded via Elementor iframe

Game 1: Nutrition Runner



- 60-second side-scrolling action game
- Collect fruits & veggies → earn points
- Dodge junk food → lose a life
- Space / ↑ / Tap · double jump
- Speed increases each level

Game 2: Nutrition Plate Builder



- MyPlate drag-and-drop game
- 6 food groups · 3 progressive rounds
- +15 pts correct · +50 pts perfect plate
- Avoid junk food — costs points
- Balance bars track each food group

7.Key Challenges & Solutions

01

Challenge

Non-standard API integration

PolyU COMP gateway uses custom endpoint (comp.azure-api.net/azure) and Api-key header auth — incompatible with standard OpenAI SDK



Solution

Custom PHP WordPress plugin

Built lightweight plugin using wp_remote_post() to construct correct headers & JSON payload. API key stored securely in WP options table.

02

Challenge

Child-safe AI content control

GPT-4o Mini responds to any topic by default — unacceptable risk on a platform designed for children aged 3–12



Solution

System prompt engineering

Curated system prompt prepended to every API call, restricting Nutri to child nutrition topics only and enforcing age-appropriate tone.

03

Challenge

Dual-audience UX design

Parents need professional credibility; children need playful engagement — early prototypes failed to satisfy both audiences simultaneously



Solution

Dual-zone design strategy

Informational pages (Home, About, Contact) use clean, professional language; interactive pages (Recipes, Game) adopt vibrant cartoon style.

8.Evaluation Results

Usability survey (n=20) + functional testing across 4 browsers · 3 device sizes

20

Participants

Parents, teachers & caregivers

4.90/5

Visual Design Score

18/20 rated 5★, 2/20 rated 4★

100%

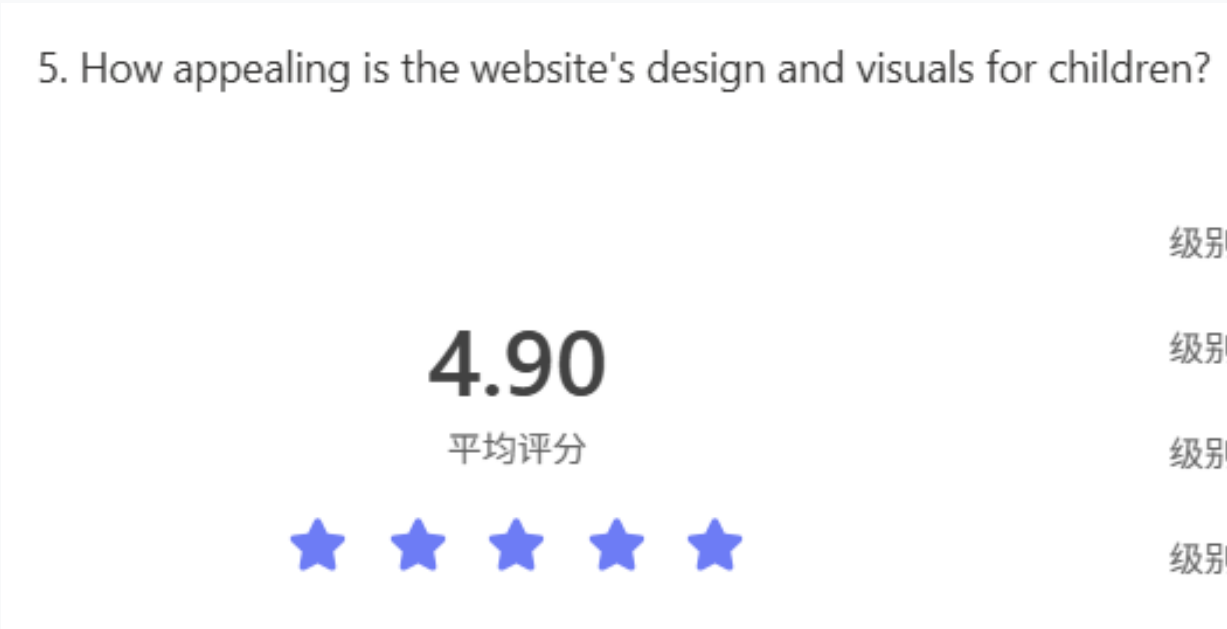
Found Nutri Helpful

All 20 answered 'Yes, very helpful'

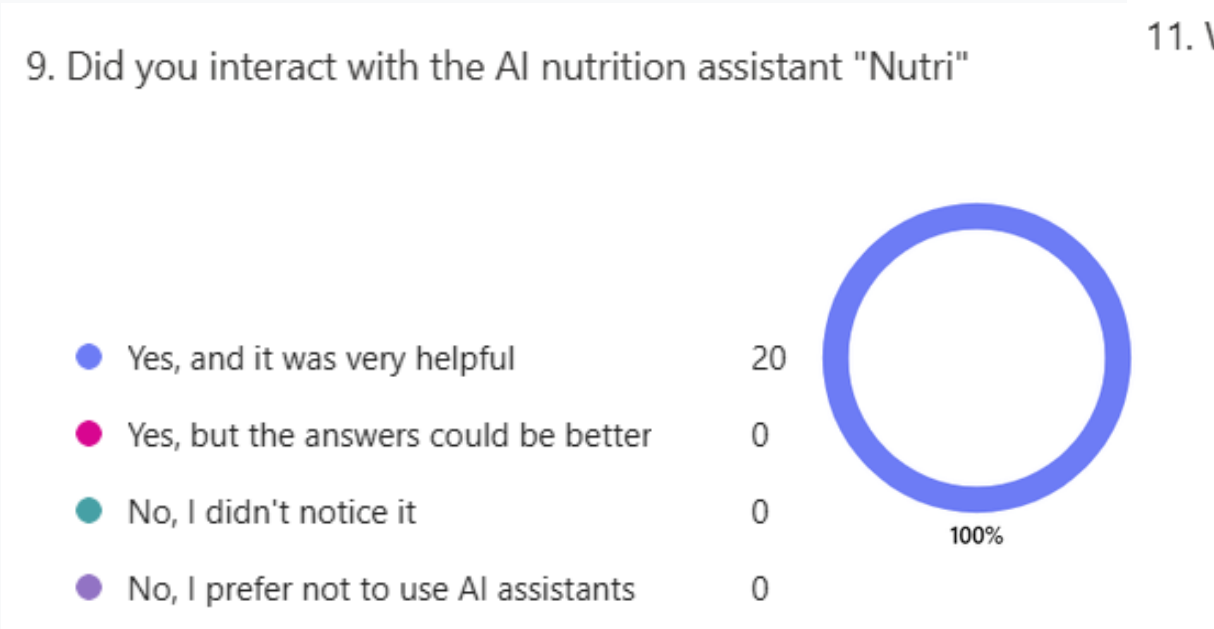
95%

Definitely Recommend

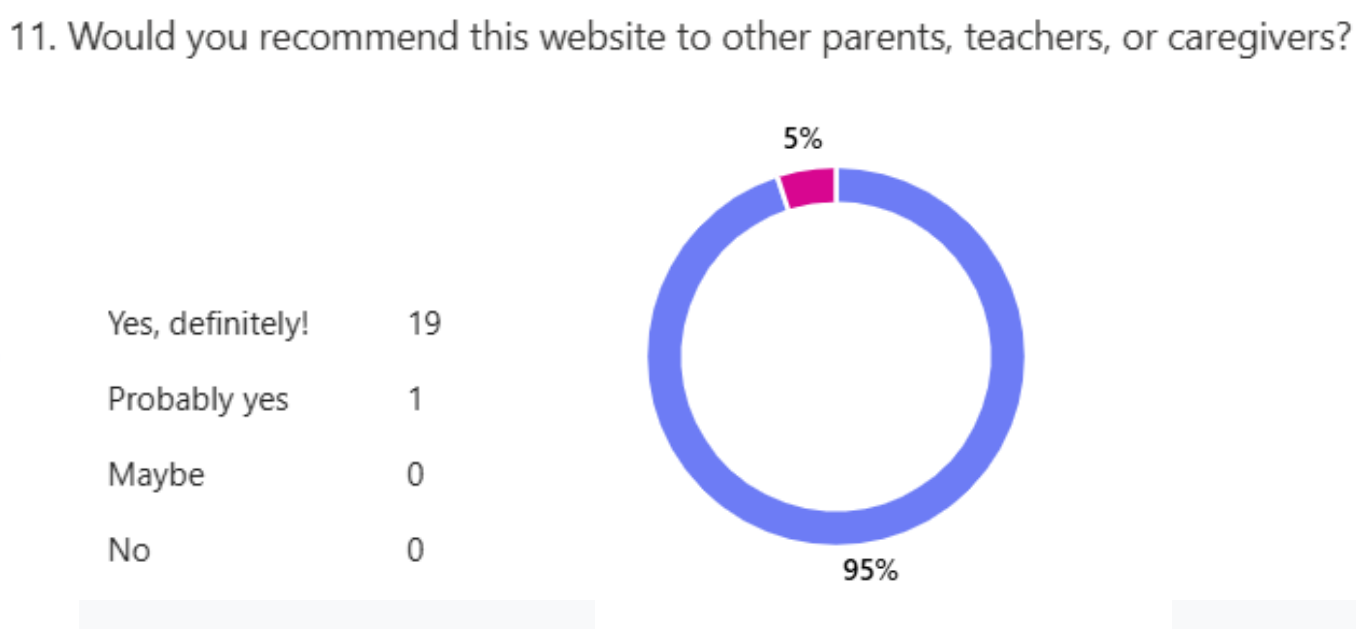
19/20 'Yes, definitely!'; 1/20 'Probably yes'



Q5: Visual design rating



Q9: Nutri AI helpfulness



Q11: Recommendation intent

9. Conclusion & Future Work

Achievements

- Nutri AI chatbot — bilingual, child-safe, live
- 50+ recipes across 7 categories — deployed
- 2 mini-games (Runner + Plate Builder) — live
- User registration system — fully functional
- All 7 modules passed testing on 4 browsers
- Usability evaluation: 4.90/5 · 100% · 95%

Contributions

- Child-safe LLM deployment via system prompt engineering
- First bilingual AI + gamification + recipe platform for children
- Reusable WordPress architecture extensible for future features

Future Work

- Image-based food logging via computer vision
- Automated recipe personalisation engine
- Direct usability testing with children aged 3–12
- Expand recipe catalog to 100+ entries

References

- [1] World Health Organization, “Healthy diet,” WHO Fact Sheet, Apr. 2020. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/healthy-diet>
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- [5] B. Huo et al., “Large language models for chatbot health advice studies: A systematic review,” *JAMA Netw. Open*, vol. 8, no. 2, p. e2457879, 2025, doi: 10.1001/jamanetworkopen.2024.57879.

Live Demo & Q&A

Demo Flow

- ① Homepage — overview & Nutri widget
- ② Recipes — browse categories & recipe detail
- ③ Nutri chatbot — EN/ZH live query
- ④ Game page — Runner + Plate Builder
- ⑤ Register page — user system

Visit the platform

kids-nutrition-fun.com

Thank you!

LIU Yibo · 22098545D · COMP4913